

**Shared GEAT engine
(Kamin et al., PRX Q 6, 020342)**

generalized entropy accumulation
for prepare-and-measure +
decoy-state QKD; min-tradeoff
functions; finite-size second order

Kamin et al. — the engine

tight finite-size key rates via
GEAT; unified decoy on Choi
states; numerical stability of
the entropy optimization.

Channel: homogeneous, not
built around public dynamic
sectors.

This work — a conditioning layer

adds a PUBLIC sector score
register $C_i = (M_i, T_i)$ on top
of the same engine; variance-
based Jensen-gap advantage
for heterogeneous DYNAMIC
channels; unified sector-
conditioned decoy + self-
certified dual minorant.

Relationship: complementary, not competing

Kamin provides the entropy-accumulation engine; this work supplies the
public sector-conditioned structure the engine consumes. The gain is over
~~global coarse-graining of a heterogeneous channel, and vanishes when the~~
channel is homogeneous.